general relativity I've ever seen) focuses solely on warped space. But it's good to bear in mind that in many common astrophysical environments, it's the warping of time that is dominant.

4. In 1974, Russell Hulse and Joseph Taylor discovered a binary pulsar system—two pulsars (rapidly spinning neutron stars) orbiting one another. Because the pulsars move very quickly and are very close together, Einstein's general relativity predicts that they will emit copious amounts of gravitational radiation. Although it is quite a challenge to detect this radiation directly, general relativity shows that the radiation should reveal itself indirectly through other means: the energy emitted via the radiation should cause the orbital period of the two pulsars to gradually decrease. The pulsars have been observed continuously since their discovery, and indeed, their orbital period has decreased—and in a manner that agrees with the prediction of general relativity to about one part in a thousand. Thus, even without direct detection of the emitted gravitational radiation, this provides strong evidence for its existence. For their discovery, Hulse and Taylor were awarded the 1993 Nobel Prize in Physics.

5. However, see note 4, above.

6. From the viewpoint of energetics, therefore, cosmic rays provide a naturally occurring accelerator that is far more powerful than any we have or will construct in the foreseeable future. The drawback is that although the particles in cosmic rays can have extremely high energies, we have no control over what slams into what—when it comes to cosmic ray collisions, we are passive observers. Furthermore, the number of cosmic ray particles with a given energy drops quickly as the energy level increases. While about 10 billion cosmic ray particles with an energy equivalent to the mass of a proton (about one-thousandth of the design capacity of the Large Hadron Collider) strike each square kilometer of earth's surface every second (and quite a few pass through your body every second as well), only about *one* of the most energetic particles (about 100 billion times the mass of a proton) would strike a given square kilometer of earth's surface each *century.* Finally, accelerators can slam particles together by making them move quickly, in opposite directions, thereby creating a large center of mass energy. Cosmic ray particles, by contrast, slam into the relatively slow moving